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PAPER

Overcoming methodological barriers in electronic nose clinical studies, a simulation data-based approach

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Supplementary material for this article is available online

Abstract

Analysis of volatile organic compounds by electronic nose (e-nose) may address gaps in non-invasive screening for neoplasia. Machine learning impacts study design and sample size requirements, but guidance on clinical study design is limited. This study evaluates how neoplasia prevalence, augmented data, and the number of e-nose devices impact sample size requirements. Simulated e-nose breath test data were created using real-world study data. We examined the effect of varying neoplasia prevalence (50%–5%) and data augmentation on model performance, as well as the impact of using multiple devices. Prediction models were developed using single value decomposition and random forest, and convolutional neural networks. Model performance was displayed as area under the receiver operating characteristics curve and F1-score. Stable model performance was defined as the phase where additional data no longer increases model performance. We found that lower neoplasia prevalence significantly increased sample size requirements, with low-prevalence settings (5%) requiring up to five times more data than high-prevalence settings (50%) for stable model performance. Model performance varied between devices, and integrating data from multiple devices required larger sample sizes. Approximately 400 data points per device at 50% prevalence, and 2100 data points at 5% prevalence, were necessary to reach stable model performance. Concluding, sample size requirements for e-nose studies are heavily influenced by disease prevalence and the number of devices used. Limiting device variability and ensuring sufficient case and control samples per device are crucial for achieving reliable predictive performance. Specific requirements will vary based on sensor and disease characteristics.

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Abbreviations

AI	artificial intelligence
ANN	artificial neural network
ASA	American Society of Anesthesiologists
AUC	area under the receiver operating characteristics curve
CNN	convolutional neural network
CRC	colorectal cancer
e-nose	electronic nose device
FIT	fecal immunochemical tests
LC	lung cancer
ML	machine learning
PCA	principal component analysis
RF	random forest
VOC	volatile organic compounds
XGBoost	extreme gradient boosting

1. Introduction

Screening and early diagnosis of malignancies potentially improves morbidity and mortality of malignancies [1]. Non-invasive methods improve accessibility and thereby effectiveness of screening programs [2]. Emerging evidence suggests that VOCs can be used as biomarker for malignancies, such as LC [3] and CRC [4–6]. This could fill the current gap in non-invasive screening methods [7]. VOCs can be detected using a low-cost, point-of-care e-nose using ML pattern recognition techniques.

Before clinical implementation, validation studies of VOC analysis using e-nose devices are imperative, due to the risk of overfitting and spurious correlations that can lead to overly optimistic claims on the accuracy [8, 9]. Ideally, this follows a multi-step approach [10]: after confirming the discriminative ability between cancer and healthy subjects in a pilot study, the accuracy can be assessed in larger study samples across the continuity of neoplastic lesions. Preferably, this encompasses larger studies to identify confounders and to provide a provisional positivity threshold.

However, when applying these steps to study designs of diagnostic tests that depend on ML, several methodological difficulties may be encountered [11]. Traditionally, studies on new diagnostic tests are designed according to the STARD [12] and TRIPOD [13] (prediction model development and validation, respectively) guidelines. For higher generalizability, large multi-center designs generate better results. However, ML prediction models are generally very data hungry, especially when correcting for background noise (e.g. VOCs in a different hospital) [14], and diversity within cohorts of participants. By using a multi-center design and multiple measuring devices, subtle data differences are introduced, resulting in more data that is required to build a prediction model. Unfortunately, the exact sample size that is needed to cover these subtle differences is difficult to determine. ML prediction models may require ten to over twenty times as many events per variable compared to conventional regression models to develop a stable AUC, due to the vast increase in parameters [14, 15]. Furthermore, an increase in number of participating centers leading to higher sample size requirements is counterintuitive to non-ML study designs, where the total study sample size is divided over the participating centers. Illustrative for this, is the abundance of pilot studies in VOC research, while large-scale validation studies are still scarce [6, 16].

The diagnostic accuracy may differ across study populations with varying neoplasia prevalence. A promising accuracy in pilot studies may not hold in follow-up studies with a lower neoplasia prevalence [17], underlining the importance of validation studies. We have previously investigated VOC analysis in exhaled breath for LC and CRC using e-nose technology. For LC, we performed a multi-center diagnostic study including 575 subjects in seven centers [18]. After a stepwise validation process [19], an AUC of 0.83 for the training set, and 0.79 in the validation set was derived. For CRC, we performed a similar study but in a CRC screening setting to mimic the setting of the anticipated application [5]. In this population, prevalence was significantly lower (5%) compared to the LC prevalence in the aforementioned study of 40%–43%, yielding an insufficient AUC of 0.54, despite promising results from a pilot study performed previously (AUC 0.76) [5].

As e-nose technology is relatively new, guidance for clinical diagnostic study designs is hardly available and demands both clinical methodological knowledge as technological knowledge. Therefore, in this study, we aim to provide insight in the effect of varying neoplasia prevalence, augmented data, and the number of measuring devices, on sample size requirements and diagnostic performance, using a series of simulation models.

2. Methods

2.1. E-nose data characteristics

We composed a simulated data set mimicking real study data from our LC- and CRC studies (table 1). We investigated the diagnostic accuracy of e-nose technology for the detection of CRC in a pilot study ($n = 447$), with a CRC prevalence of 15.7% [5]. Breath tests of patients with a histopathological diagnosis of CRC were compared to individuals with a normal colonoscopy (prevalence 28.6%) to develop a predictive model, yielding an AUC of 0.74 in the validation set. Next, we included 3469 patients over 12 study sites referred for screening colonoscopy based on a positive FIT in the Dutch CRC screening

Table 1. Overview of lung cancer study and colorectal carcinoma studies. ANN, artificial neural network; AUC, area under the receiver operating characteristics curve; CRC, colorectal carcinoma; CI, confidence interval; CNN, convolutional neural network; e-nose, electronic nose; LC, lung cancer; logres, logistic regression; ML, machine learning; RF, random forest; XGBoost, extreme gradient boosting.

	LC study [18]	CRC pilot study [5]	CRC study
Study design, sample size (n)	Multi-center cross-sectional, $n = 575$	Multi-center cross-sectional, $n = 447$	Multi-center cross-sectional, $n = 3469$
Neoplasia prevalence, n (%)	239 (41.6)	70 (15.7)	175 (5.0)
(Pre-neoplastic) lesions not included in prediction model	83 (14.4)	234 (52.4)	2425 (69.9)
Controls	253 (44.0)	128 (28.6)	869 (25.1)
Number of study sites, n	7	2	12
Number of used e-nose devices, n	5	5	21
Used ML techniques	Combined result of ANN, logres, RF, RF Extreme, XGBoost	ANN	CNN
AUC (95% CI)	0.86 (0.81–0.91)	0.74	0.54 (0.49–0.58)

program (unpublished results, clinicaltrials.gov identifier NCT03346005). In this study cohort, CRC prevalence was lower compared to the pilot study with 5.0%, and the control group with individuals without polyps or CRC had a prevalence of 25.1%.

The LC study had a sample size of 576 individuals across seven study sites, of which half was suspected of having LC, and the other half was healthy control [18]. The breath test outcomes were compared to the histopathological diagnosis of non-small cell lung carcinoma, which had a prevalence of 41.6% in the study cohort.

All breath tests were performed using the Aeonose (the eNose company, Zutphen, the Netherlands), a hand-held e-nose device using three hotplate metal-oxide sensors (AS-MLV sensors; Applied Sensors GmbH, Germany) [20]. VOCs interact with these sensors, causing redox reactions on the surface that lead to conductivity changes that can be measured. One breath profile contains over 2000 conductivity values. Unfortunately, the exact breath profile features of a disease as shown in these measurement values are as yet unclear.

2.2. Simulation data development

To analyze the effect of varying prevalence, augmented data, and number of e-nose devices on the required sample size, we used simulated data based on e-nose measurements of healthy control subjects from the LC study. On these measurements, we superimposed small deviations in the conductivity profiles, both in the time and temperature domains, to enable a distinction for positive and negative cases. These deviations were designed to mimic real breath samples. Therefore, we first aggregated multiple breath profiles into a single representative data sample by averaging across profiles. This step ensures that the resulting sample captures the overall trend while reducing random variability. Next, we introduced artificial disease signals with deviations of maximum 2% at various potential locations within the profile, accounting for the possibility that the disease may manifest inconsistently across different segments of the signal. To further simulate real-world conditions, a small 'jitter' in both time and temperature was applied at these locations, mimicking natural variations and measurement noise. Finally, we superimposed Gaussian noise across the entire data sample to simulate ambient fluctuations and external noise, thereby increasing the robustness of the model to real-world environmental variations. The superimposed deviations for the artificial disease signals are similar in size as the deviations from the Gaussian noise. As the exact signal features of CRC or LC in breath prints remain unknown, the quality of the resemblance of the simulated data was compared with the published LC studies model. To realize this, a classifying ML model was used while reducing the number of cases from the LC study. Afterwards, we compared the performance of the ML classifiers in the LC dataset with an equally sized simulated dataset. Additionally, we compared our results using the *pmsampsize* package in *R* [15], which accounts for outcome prevalence, number of predictors, and expected model performance (AUC). The number of variables was defined as equivalent to the number of features remaining after the preprocessing steps.

2.3. Prediction model development

To get from the raw sensor data to a prediction model, several steps are needed (figure 1). First, since most classifiers in ML prefer training on balanced datasets,

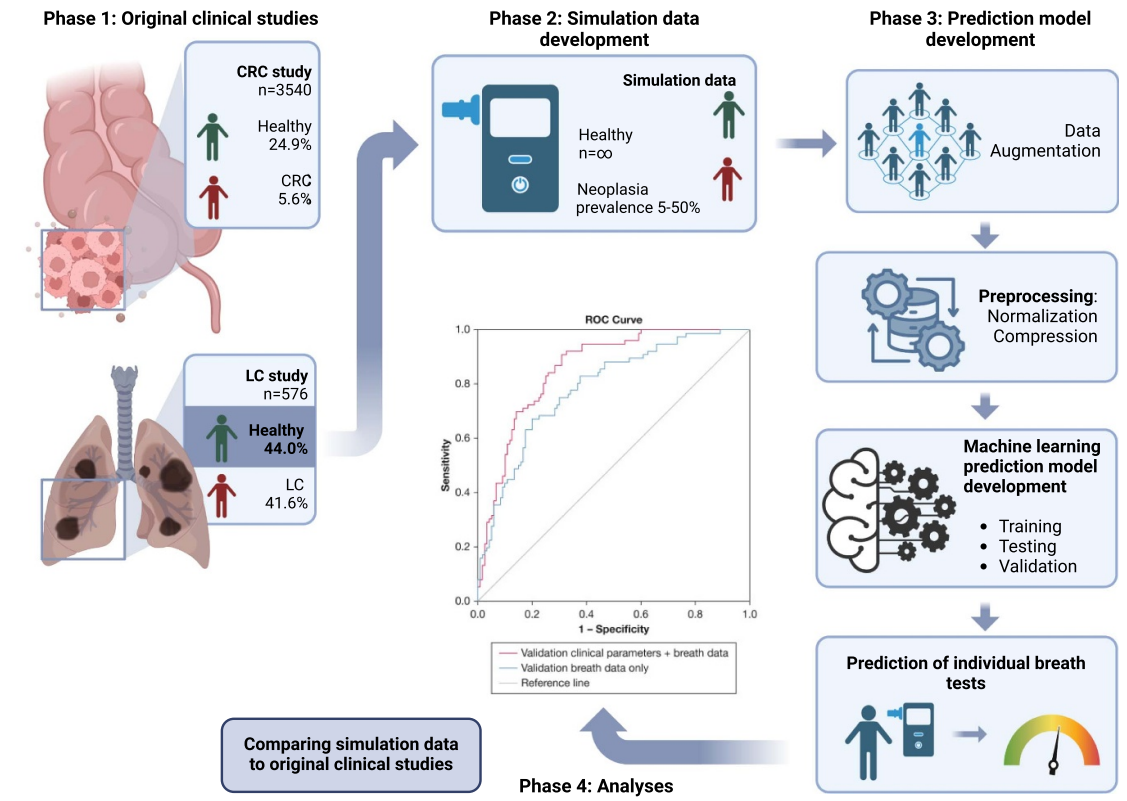

Figure 1. Graphical representation of simulation model development. We derived breath tests from healthy volunteers from our LC study (Phase 1) and superimposed small deviations to make positive and negative breath profiles with varying neoplasia prevalence values (Phase 2). To get from the raw sensor data to the prediction model (Phase 3), we first performed data augmentation using Gaussian Noise, then we preprocessed the data including standardization and compression. The pre-processed and compressed data is then used in the supervised ML model using 10% cross-validation and validation in a separate, resulting in the final prediction model to answer our research questions (Phase 4). Created with BioRender.com.

i.e. a prevalence of 50%, we used augmented data as training with a prevalence of 5% could easily lead to 'negative' classification in all cases, however still resulting in a correct outcome in 95% of all tests. Augmented data are synthetic data based on real data points, to simulate extra events to train the model [21]. In our experience, the quality of the augmented data is critical: they should not resemble the original data points too much, but they should also not be too deviant to make recognition of profiles difficult. Therefore, the parameters for generating augmented data should be chosen carefully. For our simulation data set, we used Gaussian Noise, which is a statistical noise having a probability density equal to a normal distribution [22], using the synthetic minority oversampling technique (SMOTE).

Next, to control for slight variations between multiple e-nose devices before using this data in classifying models, all data is pre-processed. For this, normalization is used by scaling the measurement amplitudes between 0 and 1. Then, as the number of conductivity values per breath profile is usually large compared to the number of participants, it is common practice to apply data compression to prevent overfitting [23]. The pre-processed and compressed data is then used in the supervised ML model. We used two different approaches for compressing and classifying both real and simulated breath profiles, using an average performance based on 10 runs of the classifier:

1. PCA as compression technique followed by a RF classifier. This approach worked well for data analysis of the LC study data. Nevertheless, a drawback of the PCA lies in its linear nature, potentially leading to the missing of non-linear features. Additionally, PCA tends to emphasize high-variance features, thereby risking the neglect of more subtle and nuanced features. We used RF with depth of 3 and 10 000 trees, proceeded by a SVD that compresses the data to 17 features.
2. Compression and classification using a CNN. The CNN is a powerful deep-learning tool, able to handle subtle differences in data sets. A disadvantage is that it requires an even larger number of data points than the PCA + RF approach. However, when using simulated data, this tool can be useful. We used a CNN consisting of the following layers:
   1. Conv2D with 8 filters, kernel size of 7, stride of 2, with padding and a relu activation.

Table 2. Overview of model performance in simulation data compared to the performed clinical studies. AUC, area under the receiver operating characteristics curve; CNN, convolutional neural network; CRC, colorectal cancer; LC, lung cancer; RFC, random forest classifier; PCA, Principal components analysis.

	Simulation	LC study [18]	Simulation	CRC pilot study [5]
Neoplasia prevalence	43%	43%	10%	15.6%
Sample size				
Training set	410	410	400	153
Validation set	135	135	135	33
AUCmax				
Training set				
CNN	99.65	84.23	99.94	82.15
PCA + RFC	84.73	84.17	x	98.68
Validation set				
CNN	82.54	73.89	72.85	73.38
PCA + RFC	50.27	76.82	x	70.0

2. Conv2D with 16 filters, kernel size of 5, stride of 2, with padding and a relu activation.
3. Batchnormalization.
4. Flatten.
5. Dense to 1 neuron with sigmoid activation.

The classifiers were selected based on experience in prior studies. To prevent overfitting, leave-10%-out cross validation is used, a resampling technique in which 10 rounds of validation are used to increase the likelihood of robust models. After a training and calibration phase, the model is tested on a blinded data set. After satisfactory performance, external validation in a separate cohort is performed.

2.4. Analyses

Using the simulated dataset, we aimed to assess three study situations. First, the effect of reducing the prevalence from an ideal 50% towards 5% (i.e. the neoplasia prevalence in the CRC study). Second, the effect of adding augmented data at different neoplasia prevalence values on the required sample size to achieve a similar model performance. Third, the effect of increasing or reducing the number of devices used in the study on the required sample size to achieve similar model performance. This was tested by comparing model performance trained on a single device to model performance trained on multiple devices.

For each situation, we calculated the effect on the model performance and plotted this against the required sample size. Performance was displayed as the F1-score and AUC of the model [24]. The F1 score is the harmonic mean of the precision (positive predicted value) and recall (sensitivity) of the model. A higher F1 score indicates a better accuracy and better representing the model accuracy in case of imbalanced datasets compared to the AUC. With this information, we aimed to get insight into sample size requirements to achieve stable model performance. Stable model was defined as the phase where additional data points no longer lead to an improvement of the AUC and F1-score.

We used Aethena (version 3.0, the eNose Company, Zutphen, the Netherlands) to generate the simulated data and Python 3.10 (Python Software foundation, Delaware, United states) and Tensorflow 2.13 (Google Brain, Alphabet Inc., California, United states) for testing the predictive model performance.

3. Results

3.1. Simulation data characteristics

The resemblance of our simulation data to real data was tested with both CNN and PCA + RF as classifier (table 2). Using PCA + RF, which were the ML techniques used for the LC study dataset, AUC values of the simulation data in the training set mimicked real data of the LC study. However, in the validation set, performance loss in the simulation dataset was greater than in the LC dataset. This performance loss is likely caused by the tendency of the PCA to prioritize high variance within the data. The artificial disease created for the simulation data is unrelated to the highest variance present in the data and is therefore 'overlooked' by PCA compression. However, by using a CNN, which combines classification and compression, we were able to reproduce the LC results (table 2). In lower prevalence settings, the model performance in the simulation dataset also approached the performance in the CRC dataset, confirming the validity of our simulation data.

3.2. Effect of prevalence

To assess the effect of neoplasia prevalence on AUC and F1-score, we used simulated data of five e-nose devices (figures 2 and S2). At decreasing prevalence from 50% (ideal training prevalence) to 5% (comparable to the CRC cohort), increasingly more data points are necessary to maintain a similar AUC. For

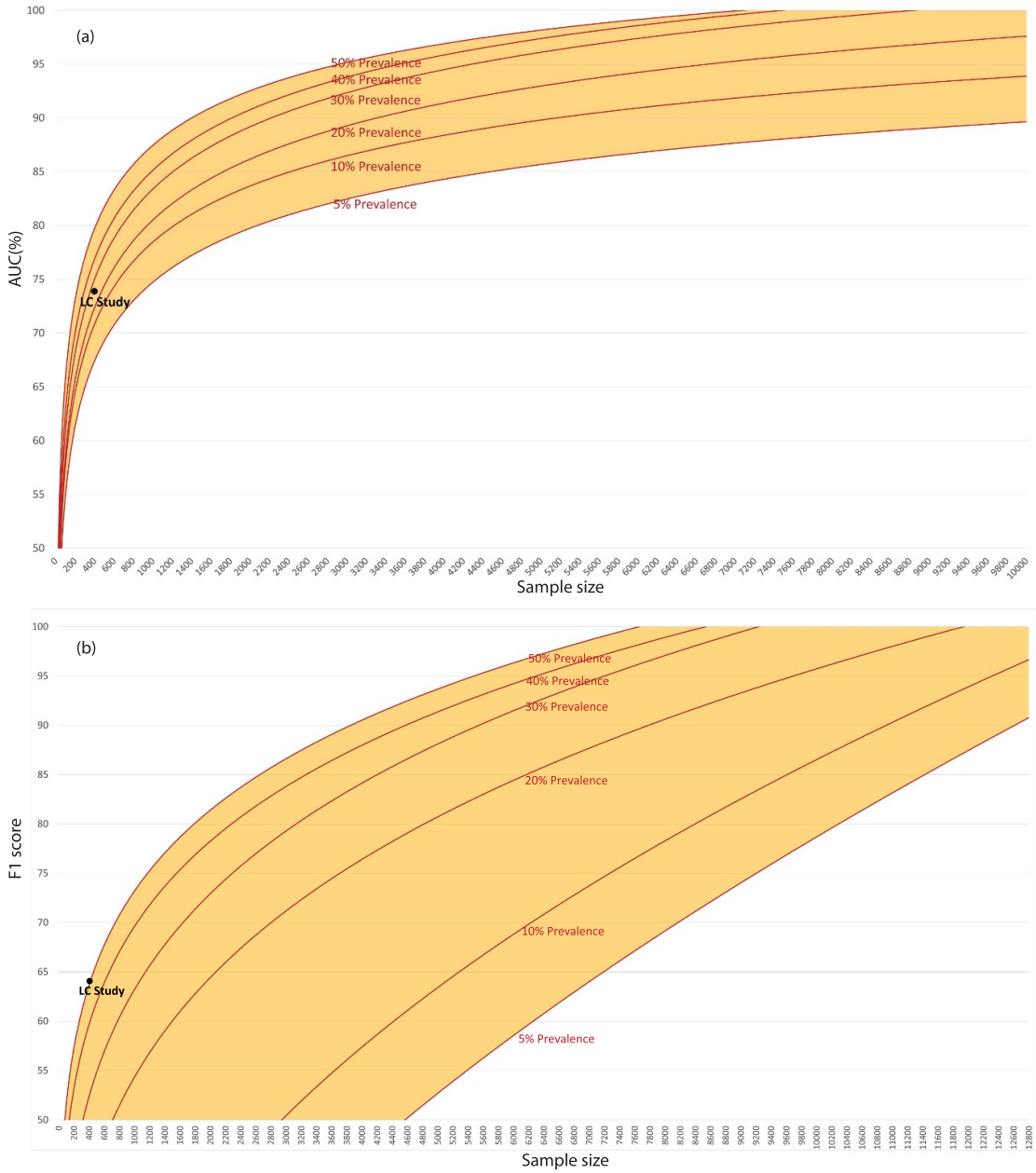

Figure 2. Effect of prevalence on required sample size on area under the curve (AUC) (A) and F1-score (B) calculated on the validation set using a convolutional neural network (CNN). For reference, the results of the LC study have been plotted in the graph. Lines represent polynomials to illustrate patterns since exact sample size requirements are related to disease and device characteristics.

example, to reach an AUC of 0.80 with a disease prevalence of 5%, approximately 2100 data points are required, compared to approximately 600 data points with a prevalence of 30% and 400 data points with a 50% prevalence. Corrected for number of positive cases at lower neoplasia prevalence (i.e. 5% and 10%), the model performance is lower as well (figure S3). Furthermore, overfitting risk is demonstrated at lower prevalence models by the loss of high training-model performance to low validation-model performance (figure S2). The risk of overfitting decreases with increasing datapoints.

3.3. Effect of augmented data

We assessed the effect of adding augmented data to the LC dataset and simulated dataset on model performance at varying sample sizes, using a CNN (figure S2, dashed lines) and an PCA + RF model (figure S4) including five e-nose devices. While augmented data may help reducing class-imbalance, the absence of model performance improvement in this figure demonstrates that augmented data is not able to substitute actual data points and provides only marginal improvement. Furthermore, when the used classifier performs insufficient,

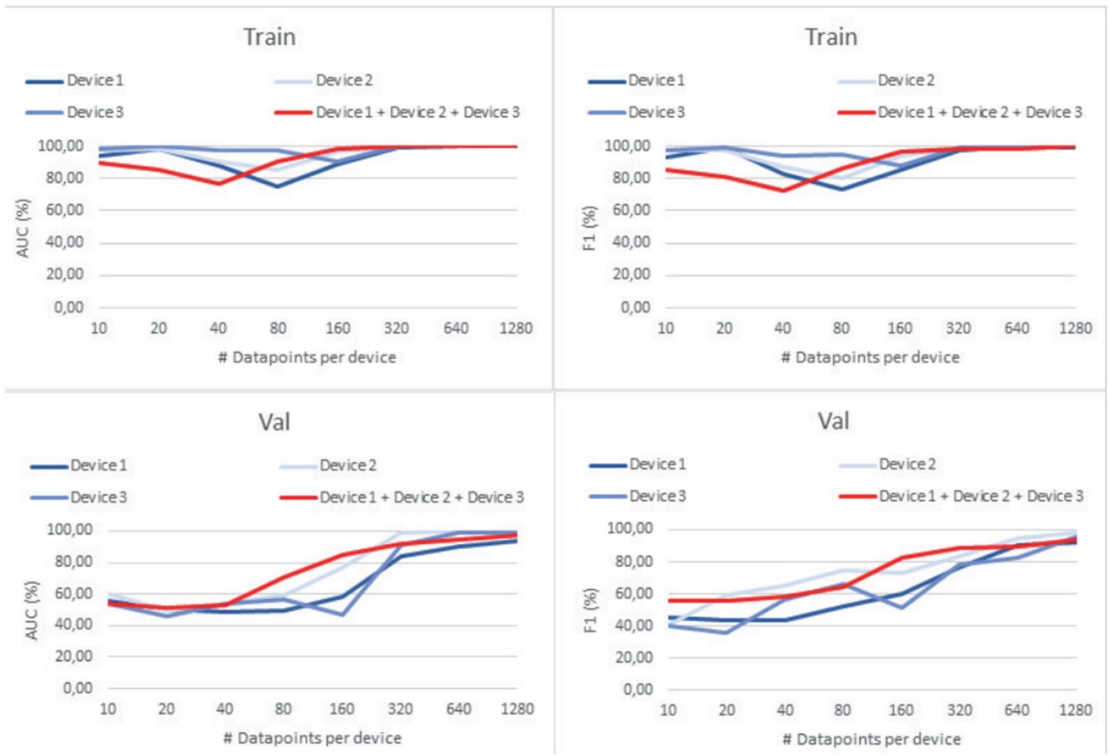

Figure 3. Effect of variation of number of datapoints per device at a 50% prevalence on model performance (AUC and F1-score) of single devices (device 1, device 2, device 3), and combination of devices (Device 1 + device 2 + device 3).

augmented data will not further improve this (figure S4).

3.4. Effect of number of devices

Model performance varies for each device (figure 3). Model training on a combination of devices does not translate into an equal increase of predictive performance. Nonetheless, model performance using combined data from three devices (i.e. model training on data from device 1, 2, and 3) was higher than the means of the model performances per individual device. Approximately 300 data points per device at a 50% prevalence are required to reach a plateau phase. Further adding devices here does not result in an additionally required total sample size.

4. Discussion

The use of VOCs as non-invasive biomarkers for screening purposes seems promising. However, data from large scale studies and external validation are lacking. This could be due to methodological difficulties in e-nose studies, including the unknown impact of background noise, varying neoplasia prevalence, and use of multiple measuring devices on sample size requirement and study design. Our results indicate that in low-prevalent settings, exhaled breath analysis may require up to five times as much data compared to high-prevalent diseases to build a stable predictive model. Augmented data only gives marginal improvement of the predictive performance. Additionally, the use of multiple study devices has a considerable impact on the sample size that is required to build an adequate predictive model.

Previous studies on sample size requirements for diagnostic test studies mainly focus on logistic regression models. Several guidelines for sample size calculation have been proposed, such as the rule of thumb of 10 events per variable [15]. However, the total sample size required is also dependent on the total number of participants, the outcome incidence, and expected model performance [15, 25]. Riley *et al* have proposed a method that includes the precision of the outcome proportion estimate, which is less likely to be overfitted [15]. Following the provided *pmsampsize* package in R, where the number of variables is set equal to the number of features that remain after preprocessing of the current studies, this results in similar outcomes as the sample size used in the published LC study [18] and our simulation data. This adds to the validity of our results.

Diagnostic performance in high-prevalent study settings in our simulation were as expected from previous pilot studies, but in low-prevalent settings, predictive performance was subpar. This correlates with findings from our CRC studies. It is known that training of ML models should be performed on a balanced dataset, i.e. number of positive cases and

controls should be present in relatively equal parts. In case of unbalanced data, it is general practice to use augmented data [21]. In contrast to reports in literature on use of augmented data in imaging techniques [21], our study results demonstrate that augmented data was not able to substitute real data points. This may explain the need for extra data points in low-prevalent study settings. A second explanation is that our simulated dataset used data of five e-nose devices to mimic actual study settings. However, our results also demonstrate that the number of devices negatively impacts the number of required positive cases, as a low number of positive cases per study device results in a low predictive performance. Together, this may clarify the effect of a low prevalence of the studied disease on the poor predictive performance in our studies.

Throughout our previous studies, multiple study devices were used, ranging from 5–6 in the LC and CRC pilot studies, to 21 in the CRC study. Due to case mix, local VOCs in ambient air, or sensor characteristics, predictive performances differed per used study device. A frequently occurring problem in e-nose devices is sensor drift over time [26]. The micro-hotplate metal-oxide sensors used in the current study device have been developed to enable transferable calibration [20], which minimizes the effect of sensor drift. Other sensor types, such as conducting polymer sensors, may be subject to more inter-device deviations. This impairs robust prediction model development because correction of sensor drift on the signals requires a large dataset of baseline breath samples [26]. Therefore, our results demonstrate that required sample size is not only dependent on number of data points and prevalence variance but may be more dependent on the number of data points per study device. Future study designs should therefore strongly consider the number of devices used, and where possible, restrict the number of devices, instead of using several devices in large multi-center studies to catch as many cases as possible in low-prevalent settings. After establishing a robust model, which is not affected by inter-device differences, external validation in a cohort using a higher number of study devices can be performed to ensure adequate performance in clinical practice.

Despite its vital importance [8, 9, 27, 28], e-nose studies with a truly external validation cohort have yet not been published. Our study demonstrates that for validation purposes in low-prevalence study settings, a substantial number of data points is required. This might explain the discrepancy between the high number of positive pilot studies and the low number of validation studies. Furthermore, this may also delay the use of e-nose technology in clinical practice in the not too far future. Given the high diagnostic performance with pooled sensitivity and specificity of over 80% in meta-analyses of pilot studies [6, 29], there is evidence for VOCs as a diagnostic biomarker, despite the variation in analysis methods. However, for true confirmation, external validation is imperative, preferably in a multi-center setting to guarantee robust results. Analysis methods of VOCs in pilot studies vary greatly, from traditional methods such as gas-chromatography-mass-spectrometry (GCMS) to the use of e-nose technology. Since traditional analysis methods such as GCMS are developed at a high calibrated level, this may result in a lower signal to noise ratio, which positively impacts sample size requirements. Clinical use of VOCs as biomarkers may therefore initially better be done with traditional methods like GCMS, albeit more labor intensive, than with e-nose devices.

Our results implicate that future study designs should consider limiting the number of study devices in a study. In addition, a minimum number of positive cases per study device should be taken into account. Since low-prevalent settings may lead to limited predictive performance, this may warrant a study design with an artificially high prevalence, over development in a setting in which testing will ultimately take place, such as for cancer screening purposes. Following this, external validation could then take place in lower prevalence settings to confirm the accuracy in the intended clinical setting.

This study has several strengths. It is the first to report on data characteristics of e-nose devices and its behavior in ML predictive models and may provide directions for future study designs. This may prevent unnecessary endeavors and associated costs for performing a study and expedite reporting of external validation studies. Although different types of e-nose devices may have proprietary components in sensors and analysis methods, the main steps are similar for different devices [23]. Our conclusions may apply beyond the specific e-nose we used during the studies.

A limitation of the use of e-nose technology is that the exact sensor signal features that are used for disease recognition remain unknown. Different types of e-nose devices all need a process of pre-processing and feature extraction to evaluate a signal with ML, and the 'black-box' characteristic of ML complicates monitoring of measured signals [23]. This inherently limits in depth analysis and more precise calculations. Additionally, it hampers designing of simulation data, which is demonstrated by the loss of predictive performance in ML models that use data compression in a separate step, unlike the CNN model that was used in our final calculations. While our simulated data was designed assuming a normal distribution, VOC distributions in breath research may not always follow a normal distribution. Testing non-normally distributed data might influence the required sample size to achieve stable model performance and should be considered in future studies. Nevertheless, the similarity of the model performance in simulation data

with both the LC study and CRC study confirms the validity of our simulation data. Additionally, different diseases, sampling media, and sampling devices may create different profiles that include more explicit features. Hypothetically, as this increases the effect size, and thereby decreases sample size requirements to achieve similar model performance, our results may not directly translate to other clinical contexts. Additionally, the use of different classifiers may be more fitting to be used in data from other e-nose devices and may lead to variations in sample size requirements to obtain stable model performance. In the future, more precise feature selection and visualization could result in more efficient data analysis, leading to fewer subjects needed to build robust disease prediction models. Therefore, a currently ongoing study aims to explore the exact features of positive and negative signals.

5. Conclusion

Sample size requirements for e-nose studies strongly depend on disease prevalence and number of study devices used throughout the study. We recommend limiting the number of devices and aim for a minimum number of 150 neoplasia cases per e-nose device for studies aiming at determining discriminative ability and a provisional positivity threshold. This may result in adapting study designs to increase the prevalence in the study cohort to be able to build a robust prediction model.

Data availability statement

No new data were created or analysed in this study.

Acknowledgment

We thank all study staff and collaborators from both clinical studies for their efforts and all subjects for their collaboration.

Ethical statement

The original studies were conducted in accordance with the declaration of Helsinki and approved by the Medical Ethical committee of the participating hospitals before study commencement. All subjects provided written informed consent.

Authors' contributions

- **Milou L.M. van Riswijk**: Conceptualization, methodology, writing of original draft, review and editing, visualization, project administration.
- **Bas van Tintelen:** Methodology, formal analysis, data curation, writing of original draft, review and editing, visualization.
- **Ruben Lucas:** Methodology, formal analysis, data curation, writing of original draft, review and editing, visualization.
- **Job van der Palen:** Conceptualization, review and editing.
- **Peter D. Siersema**: Conceptualization, review and editing, supervision.

Conflict of interest

Author P S has received research support from Pentax, The E-Nose company, Lucid Diagnostics-US, Micro Tech, Magentiq Eye, Norgine, and Endo Tools Therapeutics, and consultancy fees from Magentiq Eye. Authors M R and JvdP have no conflicts to disclose. B V T and R.L are employed by The eNose Company and are holding stock in this entity.

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